MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6BL6 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex, external cavity, pulse, and CW operation

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ik	Ehk	Pi	tk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	Vdc	W	sec	ft
Maximum:	6.8	+1.0	350	350	-700	35	+45	12	---	10,000
Minimum:	5.8	-500	---	---	-15	---	-45	---	120	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Coated unipotential
Mounting position:	Any

TEST CONDITIONS:	Ef	Ec1	Ec2	Ec3	Er
	V	Vdc	Vdc	Vdc	Vdc
	6.3	0	300	300	Adjust for max Po

GENERAL:

Qualification - Required

1/ CAUTION: Tube not designed for applications where hysteresis may affect performance.

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
4229	Total reflector current	Er = -90 Vdc (see note 3)	---	---	Ir	---	5	µAdc
1301	Heater current		---	---	If	600	750	mA
1336	Heater-cathode leakage	Ehk = ±45 Vdc	---	---	Ihk	0	75	µAdc
---	Power output (1)	See note 4						
1256	Electrode current (cathode)		---	---	Ik	15	30	mAdc
4213	Reflector voltage		---	---	Er	-30	-60	Vdc
4229	Total reflector current		---	---	Ir	---	150	µAdc
---	Frequency		---	---	F	2,110	2,170	MHz
4250	Power output		---	---	Po	25	---	mW
4214	Cathode emission	Ef = 5.7 V (see note 5)	---	---	ΔIk	---	15	%
---	Power output (2)	See note 6						
1256	Electrode current (cathode)		---	---	Ik	15	30	mAdc
4213	Reflector voltage		---	---	Er	-250	-330	Vdc
---	Frequency		---	---	F	4,225	4,355	MHz
4250	Power output		---	---	Po	25	---	mW
4214	Cathode emission	Ef = 5.7 V (see note 5)	---	---	ΔIk	---	15	%
	Quality conformance inspection, part 2							
1101	Soundness of base, base insert, or cap	See note 2	6.5	S3	---	---	---	---
1031	Low-frequency vibration	No voltages	6.5	S3	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C (see note 1)	---	---	t	500	---	hrs
---	Life-test end points							
4250	Power output (1)	See note 4	---	---	Po	20	---	mW
4250	Power output (2)	Ec2 = Ec3 = 300 Vdc (see note 6)	---	---	Po	20	---	mW

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NOTES:

1. Test tube with test cavity as specified on Drawing 166-JAN. Frequency approximately 3,000 MHz. Adjust Er for maximum Po at each life-test reading. Adjust matching network for maximum Po.

2. Water-immersion test omitted. Torque shall be applied between upper disk (g3) and molded portion of base and shall be 20 inch-pounds.

3. Static test; tube not oscillating.

4. Test tube with 2,140 MHz test cavity as specified on Drawing 166-JAN. Tube operated in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum Po. Adjust matching network for maximum Po.

5. Reduce Ef to 5.7 V while tube is oscillating. The percentage dropoff in cathode emission shall not exceed limit shown

6. Test tube with 4,290 MHz test cavity as specified on Drawing 166-JAN. Tube operated in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum Po. Adjust matching network for maximum Po.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MU
Navy -
Air Force - 11, 85
DSA - ES

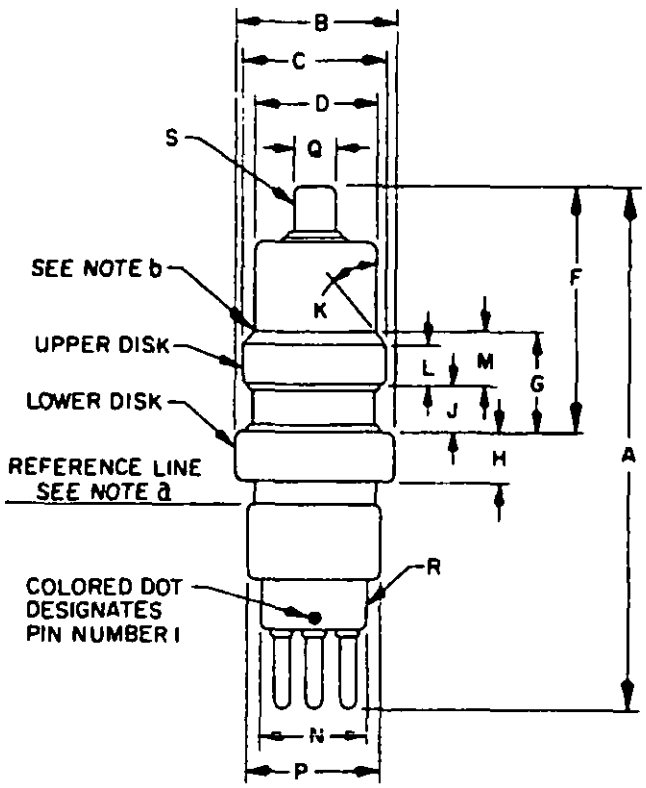
User activities:
Army - WC, ME
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

(Project 5960-2405)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	2.969 (75.41)	3.219 (81.76)
B	.990 (25.15)	1.000 (25.40)
C	.880 (22.35)	.890 (22.61)
F	1.375 (34.93)	1.500 (38.10)
M	.303 (7.70)	.323 (8.20)
Quality conformance inspection, part 2		
G	.560 (14.22)	.610 (15.49)
H	.250 (6.35)	.312 (7.92)
J	.257 (6.53)	.287 (7.29)
Quality conformance inspection, part 3 (periodic check)		
D		.750 (19.05)
K	25°	35°
N	.610 (15.49)	.656 (16.66)
P	.807 (20.50)	.822 (20.88)
Q	.245 (6.22)	.255 (6.48)
R	Base: A4-76	
S	Cap: C1-3	
Nominal dimensions		
L	.250 (6.35)	



PIN CONNECTIONS

1	g1
2	h
3	k
4	h
Lower disk	g2
Upper disk	g3
Cap	r

NOTES:
a. For dimensions below reference line refer to base A4-76.
b. Space between glass and upper disk edge or backing ring shall be .030 maximum. Quality conformance inspection, part 2 shall apply.
c. Diameters B and C shall be concentric within .025 TIR and shall form circular cylinder within a range of .006. Allowable radius at ends of cylinder shall be .030 maximum. Quality conformance inspection, part 2, shall apply.
d. Tube shall be tested in concentricity gage specified on drawing 199-JAN. Gage readings on diameters shall be as follows:

Diameter	Eccentricity
D	.050 max
N	.060 max
P	.060 max
Q	.040 max

e. All concentricity measurements shall be quality conformance inspection, part 2.
f. Pin No. 1 shall be identified by an index mark (colored dot or other).

FIGURE 1. Outline drawing of electron tube type 6BL6.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 58 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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